

4 (**short**) A tank of water of density ρ feeds two taps, at heights h_1 and h_2 below the tank's free surface, under the effect of gravity g , as shown in Fig. 1. The tank and the taps have uniform cross-sectional areas A_0 and A_1 , respectively. Friction may be neglected throughout and the flow can be considered quasi-steady.

(a) Find the two expressions for the exit velocities at the taps, V_1 and V_2 , as a function of ρ , g , h_1 , h_2 , and the velocity of the water at the free surface in the tank, V_0 . Assuming that $V_0 \ll \sqrt{2gh_1}$, simplify the above expressions and give V_1 and V_2 as a function of ρ , g , h_1 , h_2 . [6]

(b) Use mass conservation to obtain V_0 as a function of A_0 and A_1 and the previously obtained V_1 and V_2 . For a system with $g = 10 \text{ m s}^{-2}$, $h_1 = 3 \text{ m}$, $h_2 = 6 \text{ m}$, $A_0 = 0.4 \text{ m}^2$ and $A_1 = 4 \text{ cm}^2$, check if the above simplified method is justified. [4]

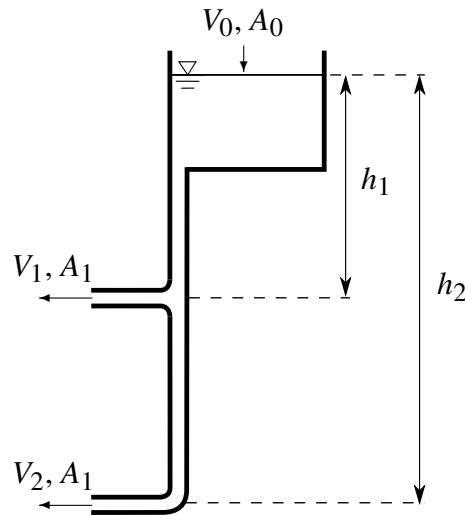


Fig. 1